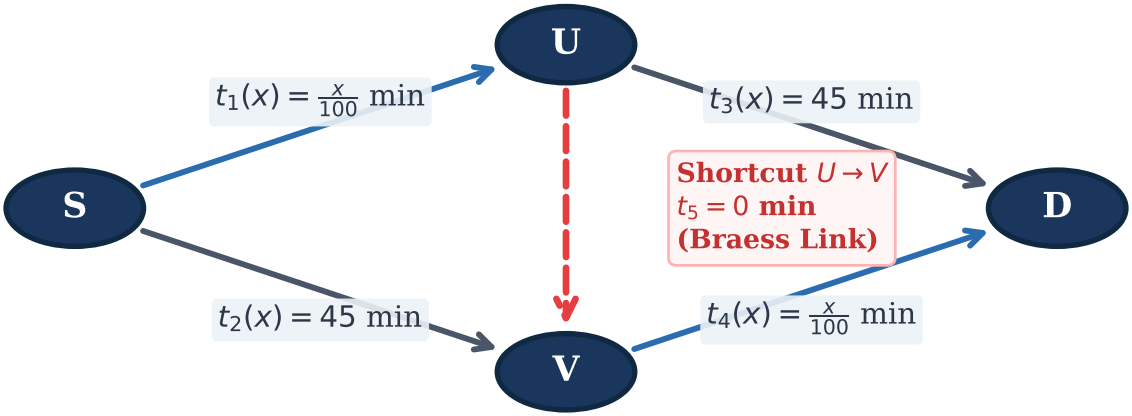
Classic Braess Paradox: Topology and Nash Equilibrium vs Social Optimum



Without Shortcut (Social Optimum):
- $x = 4000$ vehicles $\Rightarrow$ 2000 split each path
- $T_{S-U-D} = 20 + 45 = 65$ min; $T_{S-V-D} = 45 + 20 = 65$ min

With Shortcut (Selfish User Equilibrium):
- All 4000 vehicles choose $S \to U \to V \to D$
- $T = 40 + 0 + 40 =$ **80** min (+23.1% delay increase!)